Annals of Internal Medicine

CLINICAL GUIDELINE

Statin Therapy as Primary Prevention for Persons With HIV: A Synopsis of Recommendations From the U.S. Department of Health and Human Services Antiretroviral Treatment Guidelines Panel

Craig Beavers, PharmD; Alice K. Pau, PharmD; David Glidden, PhD; Emily Hyle, MD, MSc; Safia Kuriakose, PharmD; Seth S. Martin, MD, MHS; Grace McComsey, MD; Melanie Thompson, MD; Salim Virani, MD, PhD; and Jason V. Baker, MD, MS

Description: REPRIEVE (Randomized Trial to Prevent Vascular Events in HIV) showed benefits of pitavastatin as preventive therapy for atherosclerotic cardiovascular disease (ASCVD) in people with HIV (PWH). In February 2024, the U.S. Department of Health and Human Services Panel for the Use of Antiretroviral Agents in Adults and Adolescents with HIV (ARV Guidelines Panel) developed statin therapy recommendations for PWH. These recommendations were issued in collaboration with representatives from the American College of Cardiology (ACC), the American Heart Association (AHA), and the HIV Medicine Association (HIVMA). This synopsis summarizes the development process, the recommendations, and how they supplement the AHA/ACC/multisociety cholesterol guidelines and outlines gaps in primary prevention of ASCVD for PWH.

Methods: The ARV Guidelines Panel convened a writing group of 10 members (6 members of the Panel with expertise in HIV-related comorbid conditions, biostatistics, and pharmacology and 4 consultants representing ACC, AHA, and HIVMA with cardiometabolic and HIV management expertise). The writing group reviewed REPRIEVE trial data, other studies evaluating the use of statins in PWH, and the AHA/ACC/multisociety cholesterol guidelines to devise recommendations. Recommendations were based on scientific evidence with a rating scheme developed since the 1998 inception of the ARV guidelines. Proposed recommendations were presented to the full ARV Guidelines Panel, rated via vote, and approved by the Panel's voting members. These recommendations were then endorsed by ACC, AHA, and HIVMA.

Recommendations: The ARV Guidelines Panel issued a strong recommendation for initiating statin therapy among PWH with a 10-year ASCVD risk score of 5% or higher, whose absolute benefit from statins in REPRIEVE was greatest. For patients with a 10-year ASCVD risk score below 5%, the Panel favored statins but recommended patient–clinician risk discussions considering additional HIV-related factors that can increase ASCVD risk.

Ann Intern Med. doi:10.7326/ANNALS-24-03564
For author, article, and disclosure information, see end of text.
This article was published at Annals.org on 27 May 2025.

Advances in antiretroviral therapy (ART) for treatment of HIV have significantly improved life expectancy for people with HIV (PWH) over the past 3 decades. The causes of morbidity and mortality in PWH who achieve appropriate viral suppression are now dominated by conditions not related to AIDS (1, 2). Compared with people without HIV, PWH are at higher risk for many common aging-related diseases, such as atherosclerotic cardiovascular disease (ASCVD). This is likely due to increased prevalence of traditional ASCVD risk factors, prior immune depletion, inflammation caused by HIV viral persistence, toxicity from some antiretroviral (ARV) medications, higher rates of such risk factors as smoking than the general population, or some combination of these factors (1–3). Despite the clear increased risk for ASCVD, no definitive clinical data are available to inform broader or earlier use of primary ASCVD prevention strategies among PWH, compared with what is recommended in the general population. The recently completed study REPRIEVE (Randomized Trial to Prevent Vascular Events in HIV) provided high-quality clinical outcomes data to inform the use of 3-hydroxy-3-methylglutaryl coenzyme A reductase inhibitor (that is, statin) therapy as primary prevention among PWH. In response to the evidence from REPRIEVE, in February 2024, the U.S. Department of Health and Human Services Panel for the Use of Antiretroviral Agents in Adults and Adolescents with HIV (ARV Guidelines Panel) developed recommendations for use of statin therapy among PWH in collaboration with representatives from the American College of Cardiology (ACC), the American Heart Association (AHA), and the HIV Medicine Association (HIVMA) (4). This article provides a synopsis of the ARV Guidelines Panel's recommendations (**Table 1**), the process and rationale for developing these recommendations, and how they supplement the AHA/ACC/multisociety guidelines for the general population.

See also:

Editorial comment

Web-Only
Supplement

 © 2025 American College of Physicians

Table 1. ARV Guidelines Panel's Recommendations for Statin Therapy for Primary Prevention for PWH*

For PWH aged 40–75 y who have low to intermediate (<20%) 10-y ASCVD risk estimates

When 10-y ASCVD risk estimates are 5% to <20%, the ARV Guidelines Panel recommends initiating at least moderate-intensity statin therapy (AI).

Recommended options for moderate-intensity statin therapy include the following:

- Pitavastatin, 4 mg once daily (AI)
- Atorvastatin, 20 mg once daily (AII)
- Rosuvastatin, 10 mg once daily (AII)

When 10-y ASCVD risk estimates are <5%, the Panel favors initiating at least moderate-intensity statin therapy (CI). The absolute benefit from statin therapy is modest in this population; therefore, the decision to initiate statin therapy should take into account the presence or absence of HIV-related factors that can increase ASCVD risk.

Same options for moderate-intensity statin therapy as recommended for 10-y ASCVD risk estimates of 5% to <20% (see above).

For PWH aged <40 y

Data are insufficient to recommend for or against statin therapy as primary prevention of ASCVD in PWH in this age group. In the general population, lifestyle modifications are recommended, with statin therapy considered only in selected populations (see AHA)/ACC/multisociety guidelines).

Key recommendations for the general population (including PWH) based on AHA/ACC/multisociety guidelines

For people aged 40–75 y with high (≥20%) 10-y ASCVD risk estimates, initiate high-intensity statin therapy.

For people aged 20–75 y with an LDL-C level ≥4.92 mmol/L (≥190 mg/dL), initiate high-intensity statin therapy at maximum tolerated dose.

For people aged 40–75 y with diabetes mellitus, initiate at least moderate-intensity statin therapy. Perform further risk assessment to consider using a high-intensity statin.

Key considerations

Coadministration of certain statins and antiretroviral drugs may result in significant drug–drug interactions (see Table 4). In some cases, the interaction may require statin dose adjustment, switching to another statin, or increased monitoring for statin-related adverse effects.

Initiation of statin therapy should be deferred in pregnant individuals at low to intermediate ASCVD risk until after pregnancy, and statin therapy should be discontinued if a person with HIV becomes pregnant. Breastfeeding is not recommended while a person is receiving statin therapy.

ACC = American College of Cardiology; AHA = American Heart Association; ARV Guidelines Panel = U.S. Department of Health and Human Services Panel for the Use of Antiretroviral Agents in Adults and Adolescents with HIV; ASCVD = atherosclerotic cardiovascular disease; LDL-C = low-density lipoprotein cholesterol; PWH = people with HIV.

* Modified from https://clinicalinfo.hiv.gov/sites/default/files/guidelines/archive/adult-adolescent-arv-statin-therapy-2024-02-27.pdf. Rating of recommendations: A = strong, B = moderate, and C = weak. Rating of evidence: I = data from randomized controlled trials; II = data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies; and III = expert opinion.

Clinical Significance of ASCVD Among PWH

With continuous ART and viral suppression, the life expectancy of most PWH has become closer to that of people without HIV, with the estimated mortality gap now reduced to 10 years or less (1). However, compared with people without HIV, PWH have about a twofold higher risk for developing ASCVD, and their age at incident ASCVD diagnosis is about a decade younger (1, 2). Factors contributing to the development of early ASCVD include higher prevalence of traditional cardiometabolic risk factors, including obesity and smoking; incomplete immune recovery despite effective ART; toxicity of specific ARV medications; and ongoing systemic inflammation associated with HIV (2–5). Risk prediction tools for ASCVD used for the general population tend to underestimate risk among PWH (5, 6), and this underperformance is likely due to HIV disease-specific factors, such as systemic inflammation despite ART (7, 8).

Data from epidemiologic studies have established that ASCVD risk is associated with higher levels of biomarkers of inflammation among PWH than control participants without HIV (9–14). Recent data also suggest that the relative increase in ASCVD risk associated with HIV is greater among women than age-matched men, when comparing populations with and without HIV (8, 15). The mechanisms of gender- and sex-related differences in HIV-associated CVD risk broadly are an area of active investigation that may also be influenced by chronic immune activation (16, 17). Finally, structural barriers and health disparities in screening and treatment of ASCVD risk factors likely also contribute to excess ASCVD risk among PWH in the United States (18, 19).

Statins have long been posited to have advantages for PWH over other primary prevention strategies because they are associated with a well-described reduction in ASCVD events, as well as their treatment effects on inflammatory pathways (20–22). However, before the current ARV Guidelines Panel's recommendations, there were no recommendations for statin therapy as primary ASCVD prevention specifically among PWH (23–26).

AHA/ACC/Multisociety Statin Guideline for the General Population

The 2018 AHA/ACC/multisociety guideline on the management of blood cholesterol recommends a heart-healthy lifestyle across the spectrum of prevention, as well as using a risk-based approach to guide who should be considered for statin therapy (24). This risk-based approach allows clinicians to base the intensity of therapy on an individual's absolute ASCVD risk, thereby providing preventive therapies to people who are likely to benefit the most. The guidelines recommend estimation of 10-year ASCVD risk as the first

step and a patient–clinician risk discussion before initiation of statin therapy. To guide this discussion, the pooled cohort equations (PCEs) are used to estimate 10-year ASCVD risk, along with consideration of several risk enhancers not currently captured in the PCEs. Briefly, the PCEs are a validated sex- and race-specific tool based on U.S. populations derived from 5 community-based studies to estimate 10-year ASCVD risk (25). The PCEs perform well in risk estimation in a general U.S. population, with high confidence in non-Hispanic Black and White men and women. The performance of the PCEs in people from outside the United States or with varying degrees of chronic kidney disease or chronic inflammatory disease is highly variable because of the heterogeneous nature of the populations, differences in the prevalence of risk factors, and differences in population risk for ASCVD (25). Furthermore, the PCEs are not validated in those younger than 40 years.

These guidelines recommend initiation of high-intensity statin therapy for persons who are aged 40 to 75 years with a low-density lipoprotein cholesterol (LDL-C) level of at least 4.92 mmol/L (≥190 mg/dL) or a high (≥20%) 10-year ASCVD risk, and they recommend initiation of at least moderate-intensity statin therapy for those with diabetes mellitus (**Table 1**). Among adults who are aged 40 to 75 years with intermediate (7.5% to 19.9%) 10-year ASCVD risk, consideration of risk-enhancing factors (such as HIV infection) and patient–clinician discussion are recommended before initiating moderate-intensity statin therapy. The guidelines provide a comprehensive overview of all risk-enhancing factors, which are not within the scope of this review.

Methods for U.S. Department of Health and Human Services Antiretroviral Treatment Guidelines Development and Issuance of Recommendations for Statin Use in PWH

The first version, "Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents," was published in *Morbidity and Mortality Weekly Report* in 1998 (27). As advances in ART have continued to evolve over the ensuing 27 years, these guidelines have been maintained as a web-based "living guideline" with updates issued at least annually. The ARV Guidelines Panel includes a multidisciplinary team of members with expertise in HIV clinical practice and research, biostatistics, pharmacology, and related fields. All members provide a financial disclosure annually, which is publicly available on the guidelines website. **Supplement Table 1** (available at Annals.org) describes the guideline development process. Recommendations in these guidelines are based on scientific evidence and expert opinion, with a rating scheme developed since the guidelines' inception in 1998. Panel members with expertise in the topics of interest review and assess scientific evidence from published data. Each recommendation statement includes a letter (A, B, or C) that represents the strength of the recommendation and a Roman numeral (I, II, or III) that represents the quality of the evidence that supports the recommendation. **Supplement Table 2** (available at Annals.org) provides a schema of the recommendation ratings in the guidelines. These guidelines traditionally focus on the use of ART in PWH. However, because the REPRIEVE trial established the benefit of pitavastatin among PWH at low to intermediate ASCVD risk, the Panel believed it was crucial for HIV providers to have a set of recommendations on the optimal use of statins in PWH. To broaden the expertise in CVD and statin use, the ARV Guidelines Panel also invited members from ACC, AHA, and HIVMA to form a writing group of 10 members (6 from the ARV Guidelines Panel, 1 from ACC, 2 from AHA, and 1 from HIVMA). The writing group reviewed data from the REPRIEVE study (28, 29) and other studies evaluating the use of statins in HIV (30–36), as well as the AHA/ACC/multisociety cholesterol guidelines (24), to draft the recommendations and ratings. The proposed recommendations were presented to the full ARV Guidelines Panel for discussion and further revision, and the Panel's voting members ultimately voted on and approved each rated recommendation statement. These recommendations received formal endorsement from ACC, AHA, and HIVMA.

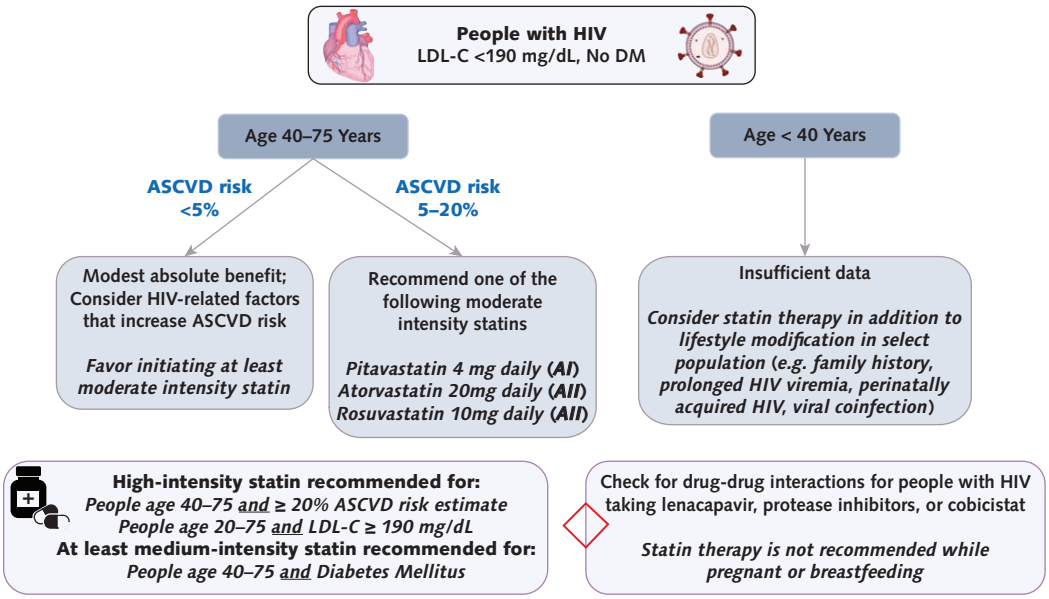
Recommendations for Statin Therapy as Primary Prevention Among Persons With HIV

Table 1 and the **Figure** summarize the ARV Guidelines Panel's recommendations and considerations to inform the use of statin therapy for primary prevention of ASCVD in PWH receiving care in the United States. The recommendations should be considered in the context of appropriate viral suppression.

Results From the REPRIEVE Study

REPRIEVE was a phase 3, global, randomized controlled trial of oral pitavastatin, 4 mg daily, versus placebo for the prevention of ASCVD in PWH aged 40 to 75 years at low to intermediate risk based on 10-year ASCVD risk estimates, as determined by the score on the AHA and ACC 2013 PCE risk calculator. Low to intermediate risk was defined as up to 15% based on LDL-C level (≥1.81 mmol/L [≥70 mg/dL]) in conjunction with LDL-C levels below specific thresholds (28, 29). People with known ASCVD were excluded, and further eligibility criteria were based on LDL-C thresholds that varied on the basis of 10-year risk estimates for ASCVD (up to a risk of 15%). The primary outcome was the occurrence of major adverse cardiac events (MACE), defined as a composite of CVD death; myocardial infarction (MI);

Figure. Recommendations for statin therapy as primary prevention among persons with HIV from the U.S. Department of Health and Human Services Panel for the Use of Antiretroviral Agents in Adults and Adolescents with HIV.



ASCVD = atherosclerotic cardiovascular disease; DM = diabetes mellitus; LDL-C = low-density lipoprotein cholesterol. Rating of recommendations: A = strong. Rating of evidence: I = data from randomized controlled trials; II = data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies. (Adapted with modifications from the ACC.org with permission from www.acc.org/Latest-in-Cardiology/Articles/2024/08/01/01/42/For-the-FITs-Closing-the-Mortality-Gap-For-People-Living-With-HIV-Updated-Recommendations-For-Statins-For-Primary-Prevention-of-ASCVD.)

hospitalization for unstable angina; stroke; transient ischemic attack; peripheral arterial ischemia; revascularization of coronary, carotid, or peripheral artery; or death of undetermined cause. A total of 7769 PWH receiving ART enrolled in the trial; 31% were female, 53% were from high-income countries (HICs), 41% were Black, and 35% were White. Participants had a median age of 50 years (IQR, 45 to 55 years), a median 10-year ASCVD risk of 4.5% (IQR, 2.1% to 7.0%), and a median CD4 T lymphocyte (CD4) cell count of 0.621×10^9 cells/L (IQR, 0.448 to 0.827×10^9 cells/L); 49% had a prior CD4 count nadir below 0.200×10^9 cells/L, and 88% had an HIV RNA level below the lower limit of quantification.

Compared with placebo, pitavastatin was associated with a 36% reduction in MACE (hazard ratio, 0.64 [95% CI, 0.48 to 0.84]), with event rates of 7.77 and 4.95 per 1000 person-years, respectively, over a median follow-up of 5.6 years. A similar treatment effect was present across the individual components of MACE. Median levels of LDL-C were 2.77 mmol/L (107 mg/dL) at baseline and 1.92 mmol/L (74 mg/dL) at month 12 in the pitavastatin group, compared with 2.75 mmol/L (106 mg/dL) at baseline and 2.72 mmol/L (105 mg/dL) at month 12 in the placebo group. Adverse events of grade 3 or higher and treatment-limiting, muscle-related symptoms occurred at higher rates in the pitavastatin group (2.4%) than in the placebo group (1.5%), as did incident diabetes mellitus (6.0% vs. 4.7%; incidence rate ratio, 1.35 [CI, 1.09 to 1.66]).

Rationale for the ARV Guidelines Panel's Recommendations According to ASCVD Risk Estimate

The overall findings from REPRIEVE, combined with the observation that equations based on traditional risk factors underestimate ASCVD risk among PWH, informed the Panel's decision to recommend the use of at least moderate-intensity statin therapy as primary prevention among PWH aged 40 to 75 years with low to intermediate 10-year ASCVD risk (<20%). To inform whether some participants received greater benefit from pitavastatin, subgroup analyses were done in REPRIEVE (27, 28). Relative risk reductions across subgroups defined by key demographic and clinical characteristics did not show a clear interaction with the treatment effect. When incident MACE was stratified by 10-year ASCVD risk score, the absolute reduction in events was greatest for people with an ASCVD risk of at least 5%. Table 2 reports the estimated number needed to treat over 5 years (NNT_5) to prevent incident MACE events with pitavastatin. Of note, the NNT_5 for people with an ASCVD risk of at least 5% was one quarter to one third that for people with an ASCVD risk less than 5%. These subgroup estimates of NNT_5 motivated the Panel to issue separate recommendations for people with ASCVD risk above and below 5%. These estimates are based on low numbers of events and should be interpreted with caution.

The relationship between a higher absolute benefit from statin therapy (that is, lower NNT_5) and higher

Table 2. NNT_5 Based on REPRIEVE*

10-Year ASCVD Risk Score, %	Participants, *n* (%)	NNT_5
>10	563 (7)	34
5–10	2995 (39)	53
2.5–<5.0	2055 (26)	130
0–<2.5	2156 (28)	187
Overall	7769 (100)	100

ASCVD = atherosclerotic cardiovascular disease; NNT_5 = number needed to treat over 5 years; REPRIEVE = Randomized Trial to Prevent Vascular Events in HIV.
* Modified from https://clinicalinfo.hiv.gov/sites/default/files/guidelines/archive/adult-adolescent-arv-statin-therapy-2024-02-27.pdf. Data from reference 29.

baseline ASCVD risk estimates is well described. This relationship should be used to inform shared decision-making discussions with PWH on the use of statins (24). Discussions on statin therapy should take into account the benefits, along with adverse effects, costs, drug–drug interaction (DDI) potential, and other patient-centered factors. The discussion should also include the increased risk for diabetes and muscle-related symptoms reported in REPRIEVE and rates of other adverse events associated with statin therapy.

The findings from REPRIEVE related to the higher absolute benefit of statin therapy among people with greater ASCVD risk motivated the Panel's decision to issue a stronger recommendation (AI) for initiating statin therapy among people with a 10-year ASCVD risk score of 5% or higher. Where the absolute benefit from statin therapy was more modest among participants with low estimated ASCVD risk (<5%), the Panel issued a weaker recommendation (CI) and highlighted that HIV-related factors not considered by the PCEs are important to consider during patient–clinician risk discussions. Using a large body of observational data, the AHA identified HIV-related factors that enhance ASCVD risk, including the following: history of prolonged HIV viremia (whether due to delayed ART initiation, nonadherence, or treatment failure), low current or nadir CD4 count ($<0.350 \times 10^9$ cells/L), lipodystrophy or lipoatrophy, metabolic syndrome, fatty liver disease, and co-infection with hepatitis C (29). Additional HIV-related factors associated with greater ASCVD risk may include but are not limited to longer duration of HIV infection and exposure to older ARV drugs with cardiometabolic toxicity (such as zidovudine, protease inhibitors, or abacavir). Conversely, people diagnosed with HIV who initiated ART with contemporary regimens (for example, integrase inhibitors) at higher CD4 counts (for example, $>500 \times 10^9$ cells/L) are likely to be at lower absolute ASCVD risk.

Use of Statins Among PWH Younger Than 40 Years

REPRIEVE did not enroll PWH younger than 40 years, and the PCEs do not provide 10-year ASCVD risk estimates for persons under 40. In the general population, lifestyle modifications to address traditional risk factors (such as diet, exercise, smoking cessation, and blood pressure control) are recommended for people younger than 40 years (24). Still, some younger persons would likely benefit from statin therapy, such as those with familial hypercholesterolemia or LDL-C levels of 4.92 mmol/L (≥190 mg/dL) or higher (in whom high-intensity statin therapy is recommended) or adults with a family history of premature ASCVD (which may also favor statin therapy) (24). Data are insufficient to inform whether risk enhancers, such as HIV-related factors, would favor statin therapy among people younger than 40 years. However, some younger PWH may be at increased ASCVD risk in the intermediate term or over their lifetime—particularly those with a very long duration of HIV infection (for example, due to perinatal exposure). Decisions on statin therapy should be individualized among this population.

Considerations on Choice of Statin Therapy for PWH

In REPRIEVE, pitavastatin was chosen in part because of less DDI potential with certain ARV medications than other statins (30, 31). Although no comparative effectiveness trials have studied various statins among PWH, studies support the treatment effect of other moderate-intensity statins for lipid lowering, reducing inflammation and immune activation, and alleviating surrogate measures of ASCVD (Table 3). Pitavastatin, atorvastatin, and rosuvastatin are all associated with greater reductions in LDL-C among PWH than pravastatin. In addition, pitavastatin (4 mg daily), rosuvastatin (10 mg daily), and high-dose atorvastatin (80 mg daily) all have shown reductions in inflammatory, monocyte, and T-cell immune activation biomarkers among PWH (32–34). Finally, in 2 separate phase 2, placebo-controlled randomized trials of statin therapy for PWH, atorvastatin (initiated at 20 mg daily) was associated with reductions in coronary noncalcified plaque by computed tomography angiography, and rosuvastatin, 10 mg daily, was associated with slower progression of common carotid artery intima–media thickness (35, 36). Cumulatively, these data motivated the ARV Guidelines Panel's recommendation for use of at least moderate-intensity statins, including pitavastatin, 4 mg daily (AI); atorvastatin, 20 mg daily (AII); or rosuvastatin, 10 mg daily (AII).

Considerations for Dosing of Statin Therapy

Statin intensity consists of 3 categories—high, moderate, and low—based on the LDL-C lowering effect (24). Table 3 outlines the statin agents and doses that qualify for each intensity based on their anticipated LDL-C lowering effects. Specifically, every reduction of 1 mmol/L (39 mg/dL) in LDL-C leads to an approximate reduction of greater than 20% in ASCVD events and 10% in all-cause mortality (36, 37).

Table 3. Intensity of Statin Therapy*

Variable	High Intensity	Moderate Intensity	Low Intensity
Estimated LDL-C lowering	≥50%	30% to 49%	<30%
Statins	Atorvastatin, 40–80 mg† Rosuvastatin, 20–40 mg†	Pitavastatin, 1–4 mg (AI)‡ Atorvastatin, 10–20 mg (AII)†‡ Rosuvastatin, 5–10 mg (AII)†‡ Fluvastatin XL, 80 mg Fluvastatin, 40 mg twice daily Lovastatin, 40–80 mg§ Pravastatin, 40–80 mg Simvastatin, 20–40 mg§	Fluvastatin, 20–40 mg Lovastatin, 20 mg§ Pravastatin, 10–20 mg Simvastatin, 10 mg§

LDL-C = low-density lipoprotein cholesterol. Rating of recommendations: A = strong. Rating of evidence: I = data from randomized controlled trials; II = data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies.

* Modified from https://clinicalinfo.hiv.gov/sites/default/files/guidelines/archive/adult-adolescent-arv-statin-therapy-2024-02-27.pdf. Data from references 28 and 29.

† Atorvastatin and rosuvastatin have drug–drug interactions with ritonavir- and cobicistat-boosted antiretroviral medications; see Table 4.

‡ These statins are included in the Panel's recommendations.

§ Simvastatin and lovastatin are contraindicated with ritonavir- and cobicistat-boosted antiretroviral medications.

However, the magnitude of LDL-C lowering varies in clinical practice.

Considerations on Drug–Drug Interactions Between Statin Therapies and ARV Medications

In current clinical practice, ART is recommended for all PWH and most commonly consists of a second-generation integrase inhibitor (such as bictegravir or dolutegravir) in combination with 2 nucleoside reverse transcriptase inhibitors (27). These medications are not expected to have significant pharmacokinetic interactions with statins. However, interactions between statins and some other ARV drugs may occur and may lead to an increase or decrease in statin drug exposure. Before having a patient start statin therapy, clinicians should consult resources to screen for DDIs and to determine which ARV drugs and which statin can be safely taken together or if dosage adjustment is necessary. Some resources with a focus on interactions between ARV drugs and statins are the Liverpool HIV Drug Interactions Checker (www.hiv-druginteractions.org/checker) (38) and the U.S. Department of Health and Human Services antiretroviral treatment guidelines (27). Because PWH may have multiple comorbid conditions and may take multiple medications, clinicians are advised to carefully review a patient's medication profile before having them start statin treatment to assess for significant interactions.

The potential for DDIs varies among the statins on the basis of their susceptibility to cytochrome P450 (CYP) enzymatic and transporter-mediated interactions. Pitavastatin, pravastatin, and rosuvastatin are not significantly metabolized by CYP enzymes. Atorvastatin, lovastatin, and simvastatin are primarily metabolized by CYP3A4, whereas fluvastatin is a major substrate of CYP2C9. All statins are susceptible to interactions related to drug transporters (for example, OAT1B1 [organic anion-transporting polypeptide 1B1]) (39).

In general, DDIs are expected between certain statins and cobicistat- or ritonavir-containing regimens, as well as the recently approved long-acting injectable medication lenacapavir. Ritonavir and cobicistat are potent inhibitors of CYP3A4 enzymes and are used as pharmacokinetic boosters to increase the drug concentrations and half-lives of some ART. The magnitude of the interaction and management strategy vary on the basis of the individual statin and ARV drugs. Lovastatin and simvastatin are contraindicated with cobicistat- or ritonavir-boosted ARV drugs, whereas no dosage adjustment is needed when these ARV drugs are used with pitavastatin. Statin dose reduction, a switch to an alternative statin, or increased monitoring for statin-related adverse effects may be necessary with atorvastatin or rosuvastatin, depending on the concomitant ARV drug (Table 4) (38).

Adverse Effects With Statin Therapy

Statin therapy is generally well tolerated and safe; however, some adverse effects have been observed. The 2018 AHA/ACC/multisociety guidelines introduced the terminology *statin-associated side effects*, as opposed to *statin intolerance*, when these adverse effects occur (24). This nomenclature was changed because most people can tolerate statin rechallenge with an alternative statin. The most frequent adverse events are statin-associated muscle symptoms, as reported in 5% to 25% of patients (40). Further, there has been no evidence of tendonitis or tendon rupture associated with statins (40).

All statins have been implicated in small increases in the risk for diabetes. The increased risk has been associated with older age and the presence of 2 or more risk factors, such as hypertension or an elevated fasting glucose level, fasting triglyceride level, or body mass index (41, 42). Pitavastatin has previously been shown to have a neutral effect on glucose levels among people with metabolic syndrome. This effect was consistent in REPRIEVE, where no treatment effect on glucose levels was observed (28, 43). However, in REPRIEVE, there was a small increase in new-onset

Table 4. Concomitant Use of ARV Drugs and Statins Recommended by the ARV Guidelines Panel as Primary ASCVD Prevention*

Panel-Recommended Statins and Doses; ARV Drugs	Recommendations
Pitavastatin, 4 mg once daily	
INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, EFV, ETR, RPV NRTI: 3TC, ABC, FTC, TAF, TDF PI/r: ATV/r, DRV/r Other: LEN, MVC	No dosage adjustment.
INSTI: EVG/c PI/c: ATV/c, DRV/c Other: FTR	No data; use standard dose and monitor for AEs.
Atorvastatin, 20 mg once daily	
INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, RPV NRTI: 3TC, ABC, FTC, TAF, TDF Other: LEN, MVC	No dosage adjustment.
INSTI: EVG/c PI: DRV/c, DRV/r	Increase atorvastatin concentrations observed. Do not exceed 20 mg/d†; monitor for AEs.
NNRTI: EFV, ETR	Decrease atorvastatin concentrations observed.
PI: ATV/c	Do not coadminister.
PI: ATV, ATV/r	Increase atorvastatin concentrations observed or possible. Monitor for AEs.
Rosuvastatin, 10 mg once daily	
INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, EFV, ETR, RPV NRTI: 3TC, ABC, FTC, TAF, TDF Other: LEN, MVC	No dosage adjustment.
INSTI: EVG/c PI: DRV/r Other: FTR	Increase rosuvastatin concentrations observed. Monitor for AEs.
PI: DRV/c	Increase rosuvastatin concentrations observed. Do not exceed 20 mg/d†; monitor for AEs.
PI: ATV, ATV/r, ATV/c	Increase rosuvastatin concentrations observed or expected. Do not exceed 10 mg/d†; monitor for AEs.

3TC = lamivudine; ABC = abacavir; AE = adverse effect; ARV = antiretroviral; ARV Guidelines Panel = U.S. Department of Health and Human Services Panel for the Use of Antiretroviral Agents in Adults and Adolescents with HIV; ASCVD = atherosclerotic cardiovascular disease; ATV = atazanavir; ATV/c = ATV-cobicistat; ATV/r = ATV-ritonavir; BIC = bictegravir; CAB = cabotegravir; DOR = doravirine; DRV/c = darunavir-cobicistat; DRV/r = DRV-ritonavir; DTG = dolutegravir; EFV = efavirenz; ETR = etravirine; EVG/c = elvitegravir-cobicistat; FTC = emtricitabine; FTR = fostemsavir; INSTI = integrase strand transfer inhibitor; LEN = lenacapavir; MVC = maraviroc; NNRTI = nonnucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PI/c = cobicistat-boosted PI; PI/r = ritonavir-boosted PI; RAL = raltegravir; RPV = rilpivirine; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate.

* Modified from https://clinicalinfo.hiv.gov/sites/default/files/guidelines/archive/adult-adolescent-arv-statin-therapy-2024-02-27.pdf. This table includes recommendations for pitavastatin, atorvastatin, and rosuvastatin when used with various ARV drugs based on potential drug–drug interactions.

† Based on recommendations from the U.S. Food and Drug Administration product label.

diabetes (6.0% in the pitavastatin group; 4.7% in the placebo group). Data are lacking on risk for new-onset diabetes with other statins among PWH, but a randomized controlled trial in PWH did show increases in insulin resistance and impaired fasting glucose levels with rosuvastatin use (44). The specific mechanisms leading to statin-associated diabetes mellitus remain unclear. Rather, it seems that a few patients with diabetic susceptibility cross the threshold to incident diabetes mellitus after statin therapy. Thus, the 2018 AHA/ACC/multisociety guidelines note that the benefits of statin therapy are shown to outweigh the risks for new-onset diabetes mellitus, and the possibility of incident diabetes mellitus should not be a contraindication to statin therapy or indication for withdrawal of statins (24).

Mild increases in liver enzyme levels are seen in some patients who receive statins; these are usually transient without clinical complications (45). Data related to any concerns about cognitive decline have been weak or contradictory and, as such, do not warrant statin avoidance or cessation (24). Before having a patient start statin therapy, clinicians should do a comprehensive evaluation of musculoskeletal symptoms; assess risk factors for diabetes; and have a clinician–patient shared decision-making discussion focused on indications, benefits, risks, and patient concerns and preferences.

Implications for Practice in Other Settings Globally

Differences in ASCVD risk and the validity of risk prediction tools have implications when considering the application of the ARV Guidelines Panel's recommendations on use of statin therapy in other settings globally. Although most PWH live in sub-Saharan Africa and other low- to middle-income countries (LMICs), risk prediction tools and prevention strategies for ASCVD have typically been validated in HICs, such as the United States and western European countries. This creates uncertainty about whether the risk for ASCVD and benefits from statin therapy that have been established in HICs can be applied in LMICs. Of note, the REPRIEVE population reflected global enrollment (although 52.7% was from HICs), with 18.3% from Latin America or the Caribbean, 7.6% from Southeast or East Asia, and 14.9% from sub-Saharan Africa (27).

The relative contribution of ASCVD as the primary manifestation of CVD, or as a major cause of morbidity and mortality among PWH, may also differ between LMICs and HICs. A systematic review describing causes of morbidity and mortality within sub-Saharan Africa during the era of ART (2008 to 2018) found that tuberculosis was the most common cause of hospitalization or death, with infectious diseases still outnumbering other noncommunicable diseases as a group (46). When the specific manifestations of CVD among PWH are considered, nonischemic causes of heart failure

account for a large proportion of CVD illness in LMICs (47, 48). Specific to HIV endemic regions within sub-Saharan Africa, important factors influencing the spectrum of individual CVD manifestations among PWH include the younger age distribution of the population (49), the higher burden of infectious diseases with CVD complications (such as tuberculosis) (50), the lower burden of dyslipidemia (51, 52), and the high prevalence of hypertension and associated hypertensive heart disease and heart failure (53). Consistent with this, data from REPRIEVE show that the rate of ASCVD events (MACE) among the placebo group was lowest in sub-Saharan Africa (1.8 per 1000 person-years) compared with other regions in the trial.

In this context, there are limitations in applying ASCVD risk scores across regions globally. A recent study compared predicted ASCVD risk using the Framingham risk score (used commonly in the United States and Canada), the PCEs (used by the AHA/ACC/multisociety cholesterol guidelines in the United States), and 4 versions of the Systematic Coronary Risk Evaluation 2 (SCORE2; recommended by the European Society of Cardiology and European Atherosclerosis Society within Europe, and the only score that considers geographic risk region) (54). This study used a simulated cohort that varied risk factor prevalence, such that this theoretical population would not directly reflect the populations used to derive or validate the risk equations. The concordance across these risk algorithms ranged between 19% and 85%, and the proportion of the population stratified specifically to moderate risk was 32% by the Framingham risk score, 39% by the PCEs, 45% by the SCORE2 version for low-risk regions, and 25% by the SCORE2 version for very-high-risk regions. The lack of validated risk scores for sub-Saharan Africa and other LMIC regions where most PWH reside greatly limits the ability to identify those who may benefit most from primary prevention strategies, such as statin therapy. Finally, recommendations on statin use in LMICs is further complicated by public health priorities competing for a smaller pool of health care resources, both in absolute dollars and when measured by the proportion of gross domestic product spent on health care.

Ongoing Clinical Questions and Priorities

Given the limitations of the PCEs both domestically and internationally, as outlined above, there have been efforts to continue to refine CVD risk estimates across all populations, including PWH. The 2018 AHA/ACC/multisociety guidelines note that other markers can help inform individual risk, including inflammatory markers, such as high-sensitivity C-reactive protein (24). Additional analyses in REPRIEVE will inform whether inflammatory biomarkers (high-sensitivity C-reactive protein and interleukin-6) measured at entry can identify participants who had a greater treatment effect from pitavastatin. In this context, it should be noted that the REPREIVE population had relatively low ASCVD risk by traditional risk factors and limited changes in high-sensitivity C-reactive protein with pitavastatin (55). One additional biomarker that may reflect residual inflammatory risk is lipoprotein(a), which has been associated with greater pericoronary inflammation among PWH than among people without HIV (56). Further analyses and additional research are needed to inform whether lipoprotein(a) or other novel inflammatory markers have a role in risk stratification for PWH.

Work is also ongoing to develop novel algorithms that will improve risk prediction. As mentioned above, several risk scores have been designed to refine estimates on the basis of global region, such as the SCORE2. Any risk score would ideally be independent of sex and race and include a younger demographic, which would allow broader generalizability. Through a science advisory group, the AHA recently developed the PREVENT (Predicting Risk of CVD EVENTs) equation to improve the estimated risk for ASCVD and heart failure separately, as well as a total CVD composite. Using more contemporary epidemiologic data, the PREVENT equation has been validated and enables 10- and 30-year risk estimates for total CVD, including heart failure, for a U.S. cohort aged 30 to 79 years, independent of race (57, 58). The score includes the variables of sex, age, cholesterol level and treatment, body mass index, systolic blood pressure and treatment, renal function, diabetes, and smoking, as well as optional models that also incorporate urinary albumin–creatinine ratio and social deprivation index as a marker of social determinants of health. Further research is needed to understand how this tool performs among PWH. In addition, research into implementation strategies, including those with a focus on removing barriers to care, will be needed.

Other future research questions to be addressed in the prevention of ASCVD among PWH include the use of lipid-modifying agents beyond statins, such as proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors, ezetimibe, inclisiran, bempedoic acid, and other agents under investigation. A recent study examined the use of evolocumab, a PCSK9 inhibitor, in PWH and dyslipidemia, which reduced LDL-C by 56.9% compared with placebo at 24 weeks (59). Although this is promising, additional confirmatory studies and clinical outcome data are needed for PCSK9 inhibitors and other novel classes of treatment.

Another important area needing further evidence and guidance is the role of type II versus type I MI in PWH younger than 40 years given the high prevalence of type II MI in this subset and the fact that statins may have a limited role in prevention of type II MI (60).

Conclusion

REPREIVE was a practice-changing clinical trial that, when combined with guidelines for the general

population, informed recommendations on use of statin therapy as primary prevention among PWH at low to intermediate ASCVD risk. The ARV Guidelines Panel issued a strong recommendation for initiating statin therapy among people with a 10-year ASCVD risk score of 5% or higher, whose absolute benefit from statins in REPRIEVE was greater. For PWH with a 10-year ASCVD risk score below 5%, the ARV Guidelines Panel favored statin therapy but also recommended that patient–clinician risk discussions consider the presence or absence of additional HIV-related factors that can increase ASCVD risk but are not included in traditional risk assessment equations. There remain important clinical questions in the management of CVD risk among PWH, including further improvements in risk stratification, as well as additional ASCVD prevention strategies focused on reducing residual risk associated with inflammation. It is important to remember that optimizing lifestyle factors (including smoking cessation, diet, and exercise) remains paramount regardless of statin use. Finally, future research is needed to better understand the absolute risk for ASCVD and other nonischemic CVD manifestations among PWH specifically within regions where most of the HIV epidemic exists globally.

From University of Kentucky College of Pharmacy, Lexington, Kentucky (C.B.); National Institutes of Health, Bethesda, Maryland (A.K.P.); University of California, San Francisco, San Francisco, California (D.G.); Massachusetts General Hospital, Boston, Massachusetts (E.H.); Frederick National Laboratory, Bethesda, Maryland (S.K.); The John Hopkins University School of Medicine, Baltimore, Maryland (S.S.M.); Case Western Reserve University, Cleveland, Ohio (G.M.); Thacker and Thompson, Atlanta, Georgia (M.T.); The Aga Khan University, Karachi, Pakistan (S.V.); and University of Minnesota, Minneapolis, Minnesota (J.V.B.).

Acknowledgment: The authors thank Ms. Marylu Schaffhauser for her support in the coordination of the work of the writing team and the U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents.

Disclosures: Disclosure forms are available with the article online.

Corresponding Author: Craig Beavers, PharmD, Department of Pharmacy Practice and Science, University of Kentucky College of Pharmacy, 783 South Limestone, Lexington, KY 40508; e-mail, cjbeav2@uky.edu.

Author contributions are available at Annals.org.

References

1. **Marcus JL, Leyden WA, Alexeeff SE, et al.** Comparison of overall and comorbidity-free life expectancy between insured adults with and without HIV infection, 2000-2016. JAMA Netw Open. 2020;3:e207954. [PMID: 32539152] doi:10.1001/jamanetworkopen.2020.7954

2. **Shah ASV, Stelzle D, Lee KK, et al.** Global burden of atherosclerotic cardiovascular disease in people living with HIV: systematic review and meta-analysis. Circulation. 2018;138:1100-1112. [PMID: 29967196] doi:10.1161/CIRCULATIONAHA.117.033369

3. **Nou E, Lo J, Hadigan C, et al.** Pathophysiology and management of cardiovascular disease in patients with HIV. Lancet Diabetes Endocrinol. 2016;4:598-610. [PMID: 26873066] doi:10.1016/S2213-8587(15)00388-5

4. **Panel on Antiretroviral Guidelines for Adults and Adolescents.** Guidelines for the Use of Antiretroviral Agents in Adults and Adolescents with HIV. U.S. Department of Health and Human Services; 2024. Accessed at https://clinicalinfo.hiv.gov/sites/default/files/guidelines/archive/adult-adolescent-arv-2024-02-27.pdf on 19 November 2024.

5. **Feinstein MJ, Nance RM, Drozd DR, et al.** Assessing and refining myocardial infarction risk estimation among patients with human immunodeficiency virus: a study by the Centers for AIDS Research Network of Integrated Clinical Systems. JAMA Cardiol. 2017;2:155-162. [PMID: 28002550] doi:10.1001/jamacardio.2016.4494

6. **Triant VA, Perez J, Regan S, et al.** Cardiovascular risk prediction functions underestimate risk in HIV infection. Circulation. 2018;137:2203-2214. [PMID: 29444987] doi:10.1161/CIRCULATIONAHA.117.028975

7. **Althoff KN, Gebo KA, Moore RD, et al; North American AIDS Cohort Collaboration on Research and Design.** Contributions of traditional and HIV-related risk factors on non-AIDS-defining cancer, myocardial infarction, and end-stage liver and renal diseases in adults with HIV in the USA and Canada: a collaboration of cohort studies. Lancet HIV. 2019;6:e93-e104. [PMID: 30683625] doi:10.1016/S2352-3018(18)30295-9

8. **Triant VA, Lee H, Hadigan C, et al.** Increased acute myocardial infarction rates and cardiovascular risk factors among patients with human immunodeficiency virus disease. J Clin Endocrinol Metab. 2007;92:2506-2512. [PMID: 17456578] doi:10.1210/jc.2006-2190

9. **Duprez DA, Neuhaus J, Kuller LH, et al; INSIGHT SMART Study Group.** Inflammation, coagulation and cardiovascular disease in HIV-infected individuals. PLoS One. 2012;7:e44454. [PMID: 22970224] doi:10.1371/journal.pone.0044454

10. **Baker JV, Hullsiek KH, Singh A, et al; CDC SUN Study Investigators.** Immunologic predictors of coronary artery calcium progression in a contemporary HIV cohort. AIDS. 2014;28:831-840. [PMID: 24370480] doi:10.1097/QAD.0000000000000145

11. **Hanna DB, Lin J, Post WS, et al.** Association of macrophage inflammation biomarkers with progression of subclinical carotid artery atherosclerosis in HIV-infected women and men. J Infect Dis. 2017;215:1352-1361. [PMID: 28199691] doi:10.1093/infdis/jix082

12. **Neuhaus J, Jacobs DR, Baker JV, et al.** Markers of inflammation, coagulation, and renal function are elevated in adults with HIV infection. J Infect Dis. 2010;201:1788-1795. [PMID: 20446848] doi:10.1086/652749

13. **Triant VA, Meigs JB, Grinspoon SK.** Association of C-reactive protein and HIV infection with acute myocardial infarction. J Acquir Immune Defic Syndr. 2009;51:268-273. [PMID: 19387353] doi:10.1097/QAI.0b013e3181a9992c

14. **So-Armah KA, Tate JP, Chang C-CH, et al; VACS Project Team.** Do biomarkers of inflammation, monocyte activation, and altered coagulation explain excess mortality between HIV infected and uninfected people? J Acquir Immune Defic Syndr. 2016;72:206-213. [PMID: 27824677] doi:10.1097/QAI.0000000000000954

15. **Baldé A, Lang S, Wagner A, et al.** Trends in the risk of myocardial infarction among HIV-1-infected individuals relative to the general population in France: impact of gender and immune status. PLoS One. 2019;14:e0210253. [PMID: 30650119] doi:10.1371/journal.pone.0210253

16. **Zanni MV, Awadalla M, Toribio M, et al.** Immune correlates of diffuse myocardial fibrosis and diastolic dysfunction among aging women with human immunodeficiency virus. J Infect Dis. 2020;221:1315-1320. [PMID: 31100122] doi:10.1093/infdis/jiz184

17. **Toribio M, Awadalla M, Cetlin M, et al.** Brief report: vascular dysfunction and monocyte activation among women with HIV. J Acquir Immune Defic Syndr. 2020;85:233-238. [PMID: 32541385] doi:10.1097/QAI.0000000000002419
18. **Richardson KK, Bokhour B, McInnes DK, et al.** Racial disparities in HIV care extend to common comorbidities: implications for implementation of interventions to reduce disparities in HIV care. J Natl Med Assoc. 2016;108:201-210.e3. [PMID: 27979005] doi:10.1016/j.jnma.2016.08.001
19. **Muiruri C, Longenecker CT, Meissner EG, et al.** Prevention of cardiovascular disease for historically marginalized racial and ethnic groups living with HIV: a narrative review of the literature. Prog Cardiovasc Dis. 2020;63:142-148. [PMID: 32057785] doi:10.1016/j.pcad.2020.02.006
20. **Ridker PM, Danielson E, Fonseca FA, et al; JUPITER Trial Study Group.** Reduction in C-reactive protein and LDL cholesterol and cardiovascular event rates after initiation of rosuvastatin: a prospective study of the JUPITER trial. Lancet. 2009;373:1175-1182. [PMID: 19329177] doi:10.1016/S0140-6736(09)60447-5
21. **Albert MA, Danielson E, Rifai N, et al; PRINCE Investigators.** Effect of statin therapy on C-reactive protein levels: the Pravastatin Inflammation/CRP Evaluation (PRINCE): a randomized trial and cohort study. JAMA. 2001;286:64-70. [PMID: 11434828] doi:10.1001/jama.286.1.64
22. **Ridker PM, Rifai N, Clearfield M, et al; Air Force/Texas Coronary Atherosclerosis Prevention Study Investigators.** Measurement of C-reactive protein for the targeting of statin therapy in the primary prevention of acute coronary events. N Engl J Med. 2001;344:1959-1965. [PMID: 11430324] doi:10.1056/NEJM200106283442601
23. **Arnett DK, Blumenthal RS, Albert MA, et al.** 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation. 2019;140:e563-e595. [PMID: 30879339] doi:10.1161/cir.0000000000000677
24. **Grundy SM, Stone NJ, Bailey AL, et al.** 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation. 2019;139:e1082-e1143. [PMID: 30586774] doi:10.1161/CIR.0000000000000625
25. **Lloyd-Jones DM, Braun LT, Ndumele CE, et al.** Use of risk assessment tools to guide decision-making in the primary prevention of atherosclerotic cardiovascular disease: a special report from the American Heart Association and American College of Cardiology. J Am Coll Cardiol. 2019;73:3153-3167. [PMID: 30423392] doi:10.1016/j.jacc.2018.11.005
26. **Mosepele M, Regan S, Massaro J, et al.** Impact of the American College of Cardiology/American Heart Association cholesterol guidelines on statin eligibility among human immunodeficiency virus-infected individuals. Open Forum Infect Dis. 2018;5:ofy326. [PMID: 30619912] doi:10.1093/ofid/ofy326
27. **Centers for Disease Control and Prevention.** Guidelines for the Use of Antiretroviral Agents in HIV-Infected Adults and Adolescents. U.S. Department of Health and Human Services; 1998. Accessed at https://clinicalinfo.hiv.gov/en/guidelines/archived-guidelines/adult-and-adolescent-guidelines on 19 November 2024.
28. **Grinspoon SK, Fitch KV, Zanni MV, et al; REPRIEVE Investigators.** Pitavastatin to prevent cardiovascular disease in HIV infection. N Engl J Med. 2023;389:687-699. [PMID: 37486775] doi:10.1056/NEJMoa2304146
29. **Grinspoon SK, Ribaudo HJ, Douglas PS.** Trial update of pitavastatin to prevent cardiovascular events in HIV infection. N Engl J Med. 2024;390:1626-1628. [PMID: 38692296] doi:10.1056/NEJMc2400870
30. **Feinstein MJ, Hsue PY, Benjamin LA, et al.** Characteristics, prevention, and management of cardiovascular disease in people living with HIV: a scientific statement from the American Heart Association. Circulation. 2019;140:e98-e124. [PMID: 31154814] doi:10.1161/CIR.0000000000000695
31. **Malvestutto CD, Ma Q, Morse GD, et al.** Lack of pharmacokinetic interactions between pitavastatin and efavirenz or darunavir/ritonavir. J Acquir Immune Defic Syndr. 2014;67:390-396. [PMID: 25202920] doi:10.1097/QAI.0000000000000333
32. **Toribio M, Fitch KV, Sanchez L, et al.** Effects of pitavastatin and pravastatin on markers of immune activation and arterial inflammation in HIV. AIDS. 2017;31:797-806. [PMID: 28252528] doi:10.1097/QAD.0000000000001427
33. **Funderburg NT, Jiang Y, Debanne SM, et al.** Rosuvastatin reduces vascular inflammation and T-cell and monocyte activation in HIV-infected subjects on antiretroviral therapy. J Acquir Immune Defic Syndr. 2015;68:396-404. [PMID: 25514794] doi:10.1097/QAI.0000000000000478
34. **Ganesan A, Crum-Cianflone N, Higgins J, et al.** High dose atorvastatin decreases cellular markers of immune activation without affecting HIV-1 RNA levels: results of a double-blind randomized placebo controlled clinical trial. J Infect Dis. 2011;203:756-764. [PMID: 21325137] doi:10.1093/infdis/jiq115
35. **Lo J, Lu MT, Ihenachor EJ, et al.** Effects of statin therapy on coronary artery plaque volume and high-risk plaque morphology in HIV-infected patients with subclinical atherosclerosis: a randomised, double-blind, placebo-controlled trial. Lancet HIV. 2015;2:e52-63-e63. [PMID: 26424461] doi:10.1016/S2352-3018(14)00032-0
36. **Longenecker CT, Sattar A, Gilkeson R, et al.** Rosuvastatin slows progression of subclinical atherosclerosis in patients with treated HIV infection. AIDS. 2016;30:2195-2203. [PMID: 27203715] doi:10.1097/QAD.0000000000001167
37. **Baigent C, Blackwell L, Emberson J, et al; Cholesterol Treatment Trialists' (CTT) Collaboration.** Efficacy and safety of more intensive lowering of LDL cholesterol: a meta-analysis of data from 170,000 participants in 26 randomised trials. Lancet. 2010;376:1670-1681. [PMID: 21067804] doi:10.1016/S0140-6736(10)61350-5
38. **University of Liverpool.** Liverpool HIV Drug Interactions Checker. Accessed at www.hiv-druginteractions.org/checker on 15 September 2024.
39. **Wiggins BS, Saseen JJ, Page RL, et al; American Heart Association Clinical Pharmacology Committee of the Council on Clinical Cardiology; Council on Hypertension; Council on Quality of Care and Outcomes Research; and Council on Functional Genomics and Translational Biology.** Recommendations for management of clinically significant drug-drug interactions with statins and select agents used in patients with cardiovascular disease: a scientific statement from the American Heart Association. Circulation. 2016;134:e468-e495. [PMID: 27754879] doi:10.1161/CIR.0000000000000456
40. **Warden BA, Guyton JR, Kovacs AC, et al.** Assessment and management of statin-associated muscle symptoms (SAMS): a clinical perspective from the National Lipid Association. J Clin Lipidol. 2023;17:19-39. [PMID: 36115813] doi:10.1016/j.jacl.2022.09.001
41. **Sattar N, Preiss D, Murray HM, et al.** Statins and risk of incident diabetes: a collaborative meta-analysis of randomised statin trials. Lancet. 2010;375:735-742. [PMID: 20167359] doi:10.1016/S0140-6736(09)61965-6
42. **Waters DD, Ho JE, Boekholdt SM, et al.** Cardiovascular event reduction versus new-onset diabetes during atorvastatin therapy: effect of baseline risk factors for diabetes. J Am Coll Cardiol. 2013;61:148-152. [PMID: 23219296] doi:10.1016/j.jacc.2012.09.042
43. **Chapman MJ, Orsoni A, Robillard P, et al.** Effect of high-dose pitavastatin on glucose homeostasis in patients at elevated risk of new-onset diabetes: insights from the CAPITAIN and PREVAIL-US studies. Curr Med Res Opin. 2014;30:775-784. [PMID: 24328357] doi:10.1185/03007995.2013.874989
44. **Erlandson KM, Jiang Y, Debanne SM, et al.** Rosuvastatin worsens insulin resistance in HIV-infected adults on antiretroviral therapy.

Clin Infect Dis. 2015;61:1566-1572. [PMID: 26157049] doi:10.1093/cid/civ554

45. **Thompson PD, Panza G, Zaleski A, et al.** Statin-associated side effects. J Am Coll Cardiol. 2016;67:2395-2410. [PMID: 27199064] doi:10.1016/j.jacc.2016.02.071

46. **Ghislain MR, Mushebenge GA, Magula N.** Cause of hospitalization and death in the antiretroviral era in sub-Saharan Africa published 2008-2018: a systematic review. Medicine (Baltimore). 2021;100:e27342. [PMID: 34713822] doi:10.1097/MD.0000000000027342

47. **Bloomfield GS, Alenezi F, Barasa FA, et al.** Human immunodeficiency virus and heart failure in low- and middle-income countries. JACC Heart Fail. 2015;3:579-590. [PMID: 26251085] doi:10.1016/j.jchf.2015.05.003

48. **Sliwa K, Carrington MJ, Becker A, et al.** Contribution of the human immunodeficiency virus/acquired immunodeficiency syndrome epidemic to de novo presentations of heart disease in the Heart of Soweto Study cohort. Eur Heart J. 2012;33:866-874. [PMID: 22048682] doi:10.1093/eurheartj/ehr398

49. **You D, Beise J, Lee S, et al.** Demographic challenges and opportunities for child health programming in Africa and Asia. BMJ. 2021;372:n19. [PMID: 33731340] doi:10.1136/bmj.n19

50. **Coates MM, Kintu A, Gupta N, et al.** Burden of non-communicable diseases from infectious causes in 2017: a modelling study. Lancet Glob Health. 2020;8:e1489-e1498. [PMID: 33098769] doi:10.1016/S2214-109X(20)30358-2

51. **So-Armah K, Benjamin LA, Bloomfield GS, et al.** HIV and cardiovascular disease. Lancet HIV. 2020;7:e279-e293. [PMID: 32243826] doi:10.1016/S2352-3018(20)30036-9

52. **Roth GA, Mensah GA, Johnson CO, et al; GBD-NHLBI-JACC Global Burden of Cardiovascular Diseases Writing Group.** Global burden of cardiovascular diseases and risk factors, 1990-2019: update from the GBD 2019 study. J Am Coll Cardiol. 2020;76:2982-3021. [PMID: 33309175] doi:10.1016/j.jacc.2020.11.010

53. **Yuyun MF, Sliwa K, Kengne AP, et al.** Cardiovascular diseases in sub-Saharan Africa compared to high-income countries: an epidemiological perspective. Glob Heart. 2020;15:15. [PMID: 32489788] doi:10.5334/gh.403

54. **Mancini GB, Ryomoto A, Yeoh E, et al.** Recommendations for statin management in primary prevention: disparities among international risk scores. Eur Heart J. 2024;45:117-128. [PMID: 37638490] doi:10.1093/eurheartj/ehad539

55. **Lu MT, Ribaudo H, Foldyna B, et al; REPRIEVE Trial Writing Group.** Effects of pitavastatin on coronary artery disease and inflammatory biomarkers in HIV: mechanistic substudy of the REPRIEVE randomized clinical trial. JAMA Cardiol. 2024;9:323-334. [PMID: 38381407] doi:10.1001/jamacardio.2023.5661

56. **Zisman E, Hossain M, Funderburg NT, et al.** Association of lipoprotein(a) with peri-coronary inflammation in persons with and without HIV infection. J Clin Lipidol. 2024;18:e430-e443. [PMID: 38403541] doi:10.1016/j.jacl.2024.02.003

57. **Khan SS, Matsushita K, Sang Y, et al; Chronic Kidney Disease Prognosis Consortium and the American Heart Association Cardiovascular-Kidney-Metabolic Science Advisory Group.** Development and validation of the American Heart Association's PREVENT equations. Circulation. 2024;149:430-449. [PMID: 37947085] doi:10.1161/CIRCULATIONAHA.123.067626

58. **Khan SS, Coresh J, Pencina MJ, et al; American Heart Association.** Novel prediction equations for absolute risk assessment of total cardiovascular disease incorporating cardiovascular-kidney-metabolic health: a scientific statement from the American Heart Association. Circulation. 2023;148:1982-2004. [PMID: 37947094] doi:10.1161/CIR.0000000000001191

59. **Boccara F, Kumar PN, Caramelli B, et al; BEIJERINCK Investigators.** Evolocumab in HIV-infected patients with dyslipidemia: primary results of the randomized, double-blind BEIJERINCK study. J Am Coll Cardiol. 2020;75:2570-2584. [PMID: 32234462] doi:10.1016/j.jacc.2020.03.025

60. **Crane HM, Paramsothy P, Drozd DR, et al; Centers for AIDS Research Network of Integrated Clinical Systems (CNICS) Cohort.** Types of myocardial infarction among human immunodeficiency virus-infected individuals in the United States. JAMA Cardiol. 2017;2:260-267. [PMID: 28052152] doi:10.1001/jamacardio.2016.5139

Author Contributions: Conception and design: C. Beavers, D. Glidden, S. Kuriakose, G. McComsey, M. Thompson, S. Virani, J.V. Baker.
Analysis and interpretation of the data: C. Beavers, D. Glidden, S.S. Martin, M. Thompson, J.V. Baker.
Drafting of the article: C. Beavers, A.K. Pau, S. Kuriakose, G. McComsey, M. Thompson, J.V. Baker.
Critical revision for important intellectual content: C. Beavers, A.K. Pau, E. Hyle, S. Kuriakose, S.S. Martin, G. McComsey, M. Thompson, S. Virani, J.V. Baker.
Final approval of the article: C. Beavers, A.K. Pau, D. Glidden, E. Hyle, S. Kuriakose, S.S. Martin, G. McComsey, M. Thompson, S. Virani, J.V. Baker.
Administrative, technical, or logistic support: C. Beavers.
Collection and assembly of data: C. Beavers, G. McComsey, J.V. Baker.